

FlightDelayDB: A Flight Delay Analysis Database System Based on Delay Factors

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1. Introduction

Flight delays are a significant issue that can lead to a chain of problems, such as disruptions to travel schedules, missed flight connections, and additional accommodation costs. These delays are particularly likely during peak seasons like holidays or vacation periods and at certain times of day when air travel demand surges, leading to airport congestion and increased chances of takeoff and landing delays. Despite this, current flight booking systems make it difficult for users to objectively assess the likelihood of delays for their specific flights. Flight delays result from a complex interplay of factors such as weather conditions, airport congestion, airline operations, aircraft conditions, and seasonal factors. However, most users choose flights based on indirect criteria such as ticket price or airline brand image, making it difficult to make informed decisions based on delay risk. This issue stems from limited access to actual flight operation factors and a lack of structured analytical data.

This project aims to systematically store flight delay factors in a database and build a system that can analyze delay trends under various conditions. By integrating flight operation information, weather forecasts, airport congestion data, and historical delay records by airline into a relational database, the system is designed to allow analysis using SQL. This enables users to look up past delay cases under similar conditions based on their flight information and estimate the likelihood of delays.

2. Background

Flight delay refers to the phenomenon where the actual departure or arrival time is later than the scheduled time, and its main causes can be categorized as follows:

Weather Factors

Wind speed, visibility, cloud ceiling, and precipitation are directly related to aircraft safety. If these exceed certain thresholds, takeoff or landing may be restricted.

Airport Congestion

When flights are concentrated at specific times, delays may occur due to runway queues, increased boarding traffic, or a shortage of gates.

Airline Operational Characteristics

Each airline's maintenance policies, aircraft turnaround time, and crew management system directly affect the likelihood of delays.

Seasonal Factors

During peak travel seasons such as holidays or vacation periods, increased demand raises the probability of delays.

Aircraft Malfunctions

Delays related to aircraft defects are influenced by the time required for maintenance and inspection.

In the United States, the Bureau of Transportation Statistics (BTS) provides structured flight delay data categorized by airline, time period, and airport. This enables a wide range of analytical studies. In contrast, in Korea, delay information is partially available, but lacks a comprehensive dataset like that of BTS, making it difficult to analyze specific delay factors in detail. This highlights the need for a database system that can systematically store and analyze flight delay data.

3. Design and Implementation

This chapter describes the database design process and implementation details.

3.1 Database Design

The database in this project is designed to store various factors that affect flight delays. It is composed of the following main entities:

Airline, Airport, Aircraft

Seasonality

Weather, Weather_Forecast

Flight

Flight_Weather, Flight_Forecast

Delay_Reason, Flight_Delay

AirportCongestion

User, User_Ticket, Prediction_Result

3.1.1 ER Model

The ER model is designed to center around the Flight entity, with weather information, seasonal data, airport congestion, airline information, and delay records organically connected. This allows for delay analysis based not on a single factor but on a combination of factors. The figure below shows the ER model for this project.



3.1.2 Description of Key Entities

Airline, Airport, Aircraft

These are components of a flight. To prevent duplication, UNIQUE constraints are applied to the IATA code (for airlines), airport code, and aircraft registration number.

Seasonality

This stores information about peak and off-peak seasons, enabling the analysis of delay trends based on seasonality.

Weather / Weather_Forecast

Stores both actual and forecasted weather data. The forecast table includes a severity score to enhance the precision of the analysis.

Flight

This is the core table that stores both scheduled and actual flight data. It is linked to seasonal and airport information and serves as the basis for delay analysis.

Flight_Weather, Flight_Weather_Forecast

These are N:M mapping tables that connect flights with weather data.

Delay_Reason, Flight_Delay

These tables record the reasons for delays and the actual delay durations. The delay_type field distinguishes between departure and arrival delays.

AirportCongestion

Stores congestion levels at airports by time slots, allowing analysis of how congestion contributes to delays.

User, User_Ticket, Prediction_Result

Stores information about users, their tickets, and prediction results, providing a foundation for personalized delay analysis.

3.2 Implementation

3.2.1 Database Creation

```
1 DROP DATABASE IF EXISTS flight_delay_db;
2
3 CREATE DATABASE flight_delay_db
4     DEFAULT CHARACTER SET utf8mb4
5     DEFAULT COLLATE utf8mb4_general_ci;
6
7 USE flight_delay_db;
8
9
```

3.2.2 Foreign Key Initialization



```
1 SET FOREIGN_KEY_CHECKS = 0;
2
3 TRUNCATE TABLE Prediction_Result;
4 TRUNCATE TABLE User_Ticket;
5 TRUNCATE TABLE User;
6 TRUNCATE TABLE Flight_Delay;
7 TRUNCATE TABLE Delay_Reason;
8 TRUNCATE TABLE Flight_Weather_Forecast;
9 TRUNCATE TABLE Flight_Weather;
10 TRUNCATE TABLE Weather_Forecast;
11 TRUNCATE TABLE Weather;
12 TRUNCATE TABLE AirportCongestion;
13 TRUNCATE TABLE Flight;
14 TRUNCATE TABLE Aircraft;
15 TRUNCATE TABLE Seasonality;
16 TRUNCATE TABLE Airport;
17 TRUNCATE TABLE Airline;
18
19 SET FOREIGN_KEY_CHECKS = 1;
20
```

3.2.3 Table Creation SQL

```

1 -- 6. DATABASE 00/01/01
2 -- 6. DATABASE 00/01/01
3 -- 6. DATABASE 00/01/01
4 DROP DATABASE IF EXISTS flight_delay_db;
5 CREATE DATABASE flight_delay_db;
6 USE flight_delay_db;
7
8 -- 1. Airline / Airport / Aircraft
9 -- 1. Airline / Airport / Aircraft
10 -- 1. Airline / Airport / Aircraft
11
12 CREATE TABLE Airline (
13     airline_id INT AUTO_INCREMENT PRIMARY KEY,
14     airline_name VARCHAR(100) NOT NULL,
15     iata_code VARCHAR(3) NOT NULL,
16     country VARCHAR(50) NOT NULL,
17     UNIQUE (iata_code)
18 );
19
20 CREATE TABLE Airport (
21     airport_id INT AUTO_INCREMENT PRIMARY KEY,
22     airport_code VARCHAR(10) NOT NULL,
23     name VARCHAR(100) NOT NULL,
24     city VARCHAR(50),
25     country VARCHAR(50) NOT NULL,
26     UNIQUE (airport_code)
27 );
28
29 CREATE TABLE Aircraft (
30     aircraft_id INT AUTO_INCREMENT PRIMARY KEY,
31     tail_number VARCHAR(50) NOT NULL,
32     model VARCHAR(50) NOT NULL,
33     manufacture_year INT,
34     airline_id INT NOT NULL,
35     UNIQUE (tail_number),
36     FOREIGN KEY (airline_id) REFERENCES Airline(airline_id)
37 );
38
39 -- 2. Seasonality (00/01/01)
40 -- 2. Seasonality (00/01/01)
41 -- 2. Seasonality (00/01/01)
42
43 CREATE TABLE Seasonality (
44     season_id INT AUTO_INCREMENT PRIMARY KEY,
45     season_name VARCHAR(100) NOT NULL,
46     start_date DATE NOT NULL,
47     end_date DATE NOT NULL,
48     season_type ENUM('PEAK', 'OFF_PEAK', 'NORMAL') NOT NULL,
49     UNIQUE (season_name)
50 );
51
52 -- 3. Weather (00/01/01)
53 -- 3. Weather (00/01/01)
54 -- 3. Weather (00/01/01)
55
56 CREATE TABLE Weather (
57     weather_id INT AUTO_INCREMENT PRIMARY KEY,
58     airport_id INT NOT NULL,
59     weather_datetime DATETIME NOT NULL,
60     temperature FLOAT,
61     wind_speed_kt FLOAT,
62     visibility_km FLOAT,
63     precipitation FLOAT,
64     weather_condition VARCHAR(50),
65     FOREIGN KEY (airport_id) REFERENCES Airport(airport_id)
66 );
67
68 -- 4. Weather Forecast (00/01/01)
69 -- 4. Weather Forecast (00/01/01)
70 -- 4. Weather Forecast (00/01/01)
71
72 CREATE TABLE Weather_Forecast (
73     forecast_id INT AUTO_INCREMENT PRIMARY KEY,
74     airport_id INT NOT NULL,
75     forecast_datetime DATETIME NOT NULL,
76     predicted_at DATETIME NOT NULL,
77     weather_condition VARCHAR(50),
78     temperature FLOAT,
79     wind_direction INT,
80     wind_speed_kt FLOAT,
81     cloud_ceiling_ft INT,
82     visibility_km FLOAT,
83     crossing_h FLOAT,
84     confidence_score FLOAT,
85
86     -- 00/01/01 (0-1)
87     severity_score FLOAT,
88     FOREIGN KEY (airport_id) REFERENCES Airport(airport_id)
89 );
90
91 -- 5. Flight (00/01/01)
92 -- 5. Flight (00/01/01)
93 -- 5. Flight (00/01/01)
94
95 CREATE TABLE Flight (
96     flight_id INT AUTO_INCREMENT PRIMARY KEY,
97     airline_id INT NOT NULL,
98     aircraft_id INT NOT NULL,
99     flight_number VARCHAR(20) NOT NULL,
100     departure_airport INT NOT NULL,
101     arrival_airport INT NOT NULL,
102     scheduled_dep DATETIME NOT NULL,
103     scheduled_arr DATETIME,
104     estimated_dep DATETIME,
105     actual_dep DATETIME,
106     status VARCHAR(20),
107
108     -- 00/01/01 + 00/01/01
109     season_id INT NOT NULL,
110     arrival_season_id INT NOT NULL,
111
112     FOREIGN KEY (airline_id) REFERENCES Airline(airline_id),
113     FOREIGN KEY (aircraft_id) REFERENCES Aircraft(aircraft_id),
114     FOREIGN KEY (departure_airport) REFERENCES Airport(airport_id),
115     FOREIGN KEY (arrival_airport) REFERENCES Airport(airport_id),
116     FOREIGN KEY (season_id) REFERENCES Seasonality(season_id),
117     FOREIGN KEY (arrival_season_id) REFERENCES Seasonality(season_id),
118     UNIQUE (flight_number, scheduled_dep, departure_airport)
119 );
120
121 -- 6. Flight - Weather (00/01/01)
122 -- 6. Flight - Weather (00/01/01)
123 -- 6. Flight - Weather (00/01/01)
124
125 CREATE TABLE Flight_Weather (
126     id INT AUTO_INCREMENT PRIMARY KEY,
127     flight_id INT NOT NULL,
128     weather_id INT NOT NULL,
129     FOREIGN KEY (flight_id) REFERENCES Flight(flight_id),
130     FOREIGN KEY (weather_id) REFERENCES Weather(weather_id),
131     UNIQUE (flight_id, weather_id)
132 );
133
134 CREATE TABLE Flight_Weather_Forecast (
135     id INT AUTO_INCREMENT PRIMARY KEY,
136     flight_id INT NOT NULL,
137     forecast_id INT NOT NULL,
138     FOREIGN KEY (flight_id) REFERENCES Flight(flight_id),
139     FOREIGN KEY (forecast_id) REFERENCES Weather_Forecast(forecast_id),
140     UNIQUE (flight_id, forecast_id)
141 );
142
143 -- 7. Delay Reason + Flight Delay
144 -- 7. Delay Reason + Flight Delay
145 -- 7. Delay Reason + Flight Delay
146
147 CREATE TABLE Delay_Reason (
148     reason_id INT AUTO_INCREMENT PRIMARY KEY,
149     name VARCHAR(50) NOT NULL,
150     category VARCHAR(50) NOT NULL,
151     UNIQUE (name)
152 );
153
154 CREATE TABLE Flight_Delay (
155     delay_id INT AUTO_INCREMENT PRIMARY KEY,
156     flight_id INT NOT NULL,
157     delay_minutes INT NOT NULL,
158     delay_reason_id INT NOT NULL,
159     reported_time DATETIME,
160
161     -- 00/01/01 + 00/01/01
162     delay_type ENUM('DEPARTURE', 'ARRIVAL', 'GENERAL') DEFAULT 'GENERAL',
163     FOREIGN KEY (flight_id) REFERENCES Flight(flight_id),
164     FOREIGN KEY (delay_reason_id) REFERENCES Delay_Reason(reason_id)
165 );
166
167 -- 8. Airport Congestion
168 -- 8. Airport Congestion
169 -- 8. Airport Congestion
170
171 CREATE TABLE AirportCongestion (
172     congestion_id INT AUTO_INCREMENT PRIMARY KEY,
173     airport_id INT NOT NULL,
174     timeslot DATETIME NOT NULL,
175     total_flights INT,
176     congestion_level ENUM('LOW', 'MEDIUM', 'HIGH') NOT NULL,
177     FOREIGN KEY (airport_id) REFERENCES Airport(airport_id),
178     UNIQUE (airport_id, timeslot)
179 );
180
181 -- 9. User / Ticket / Prediction
182 -- 9. User / Ticket / Prediction
183 -- 9. User / Ticket / Prediction
184
185 CREATE TABLE User (
186     user_id INT AUTO_INCREMENT PRIMARY KEY,
187     name VARCHAR(50) NOT NULL,
188     email VARCHAR(100) NOT NULL,
189     phone_number VARCHAR(20),
190     birthdate DATE,
191     nationality VARCHAR(50),
192     passport_number VARCHAR(50),
193     UNIQUE (email)
194 );
195
196 CREATE TABLE User_Ticket (
197     ticket_id INT AUTO_INCREMENT PRIMARY KEY,
198     user_id INT NOT NULL,
199     flight_id INT NOT NULL,
200     booking_date DATE NOT NULL,
201     seat_number VARCHAR(10),
202     FOREIGN KEY (user_id) REFERENCES User(user_id),
203     FOREIGN KEY (flight_id) REFERENCES Flight(flight_id),
204     UNIQUE (user_id, flight_id, seat_number)
205 );
206
207 CREATE TABLE Prediction_Result (
208     prediction_id INT AUTO_INCREMENT PRIMARY KEY,
209     ticket_id INT NOT NULL,
210     predicted_delay INT NOT NULL,
211     predicted_date DATETIME NOT NULL,
212     confidence_score FLOAT,
213     FOREIGN KEY (ticket_id) REFERENCES User_Ticket(ticket_id)
214 );
215
216 -- 10. Index
217 -- 10. Index
218 -- 10. Index
219
220 CREATE INDEX idx_flight_dep_time ON Flight (departure_airport, scheduled_dep);
221 CREATE INDEX idx_weather_forecast_time ON Weather_Forecast (airport_id, forecast_datetime);
222 CREATE INDEX idx_flight_weather_flight ON Flight_Weather (flight_id);
223 CREATE INDEX idx_flight_weather_forecast ON Flight_Weather_Forecast (flight_id);
224 CREATE INDEX idx_flight_weather ON Flight_Weather (flight_id, weather_id);
225 CREATE INDEX idx_flight_weather_forecast ON Flight_Weather_Forecast (flight_id, forecast_id);
226 CREATE INDEX idx_flight_weather_forecast ON Flight_Weather_Forecast (flight_id, forecast_id);
227 CREATE INDEX idx_flight_weather_forecast ON Flight_Weather_Forecast (flight_id, forecast_id);
228 CREATE INDEX idx_flight_weather_forecast ON Flight_Weather_Forecast (flight_id, forecast_id);
229
230

```

3.2.4 Integrity and Normalization Verification

To ensure that the designed database structure complies with the core constraints of the relational model, integrity and normalization checks were performed. The validation focused on the following aspects:

UNIQUE Constraint Verification

Verified that attributes that must not be duplicated — such as airline IATA codes, airport codes, and user emails — are properly protected through unique constraints.

Foreign Key (FK) Referential Integrity Verification

Checked that the Flight table correctly references higher-level entities like Airline, Airport, and Aircraft. Also verified that the DBMS throws errors when non-existent keys are inserted.

Mapping Table N:M Integrity Verification

Ensured that join tables like Flight_Weather_Forecast prevent the insertion of duplicate combinations through appropriate constraints.

Normalization Structure Verification

Reviewed all tables to ensure they adhere to the Third Normal Form (3NF), minimizing data redundancy and preventing update anomalies.

3.2.5 Test Results

This section presents the results of tests conducted to verify whether the constraints explained in 3.2.4 were properly enforced during actual data manipulation. Tests were carried out by executing SQL statements and observing DBMS responses.

(1) Foreign Key (FK) Constraint Test

When attempting to insert a Flight record with a non-existent airline ID, the DBMS returned a referential integrity violation error, confirming proper enforcement.

```
1 INSERT INTO Flight (  
2     airline_id, aircraft_id, flight_number,  
3     departure_airport, arrival_airport,  
4     scheduled_dep  
5 ) VALUES (  
6     999,      -- 존재하지 않는 항공사 ID  
7     NULL, 'TEST001',  
8     1, 2,  
9     '2025-12-01 10:00'  
10 );  
11
```

result

```
46 16:45:21 INSERT INTO Flight ( airline_id, aircraft_id, flight_number, departure_airport, arrival_airport, scheduled_dep, ... Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails ('flight_delay_db'.flight', CON...
```

(2) UNIQUE Constraint Test

When attempting to insert a duplicate email address into the User table, the DBMS returned a

duplicate key error, confirming the UNIQUE constraint worked as intended.

```
1 INSERT INTO User (name, email) VALUES ('A', 'dup@example.com');
2 INSERT INTO User (name, email) VALUES ('B', 'dup@example.com'); -- 실패해야 함
```

result

```
52 16:47:14 INSERT INTO User (name, email) VALUES ('Test User 2', 'dup@example.com') Error Code: 1062. Duplicate entry 'dup@example.com' for key 'user.email'
53 16:47:16 INSERT INTO User (name, email) VALUES ('Test User 1', 'dup@example.com') Error Code: 1062. Duplicate entry 'dup@example.com' for key 'user.email'
```

(3) Mapping Table Duplicate Insertion Test

Inserting a duplicate flight_id-forecast_id pair into the Flight_Weather_Forecast table triggered a UNIQUE constraint violation, preventing redundant data.

```
1 INSERT INTO Flight_Weather_Forecast (flight_id, forecast_id)
2 VALUES (1, 1);
3
4 INSERT INTO Flight_Weather_Forecast (flight_id, forecast_id)
5 VALUES (1, 1); -- 중복 삽입
6
7
```

result

```
282 19:41:46 INSERT INTO Flight_Weather_Forecast (flight_id, forecast_id) VALUES (1, 1) Error Code: 1062. Duplicate entry '1-1' for key 'flight_weather_forecast_flight_id'
```

(4) Sample Data Insertion for Functional Testing

Sample data covering delay patterns per airline, weather conditions, seasonal periods, time slots, and congestion levels was inserted to test the system's full analysis pipeline. The system

operated as expected, indicating that the analysis functions were working properly.

```
1 USE flight_delay_db;
2
3 -- -----
4 -- 1. Delay Reason
5 -- -----
6 INSERT INTO Delay_Reason (name, category)
7 VALUES
8 ('기상', 'Weather'),
9 ('기계적결함', 'Mechanical'),
10 ('관제', 'ATC'),
11 ('운행스케줄', 'Operational');
12
13
14 -- -----
15 -- 2. Airline
16 -- -----
17 INSERT INTO Airline (airline_name, iata_code, country)
18 VALUES
19 ('Korean Air', 'KE', 'South Korea'),
20 ('Asiana Airlines', 'OZ', 'South Korea'),
21 ('Delta Airlines', 'DL', 'USA');
22
23
24 -- -----
25 -- 3. Airport
26 -- -----
27 INSERT INTO Airport (airport_code, name, city, country)
28 VALUES
29 ('ICN', 'Incheon Intl Airport', 'Incheon', 'South Korea'),
30 ('NRT', 'Narita Airport', 'Tokyo', 'Japan'),
31 ('LAX', 'Los Angeles Intl', 'Los Angeles', 'USA');
32
33
34 -- -----
35 -- 4. Seasonality
36 -- -----
37 INSERT INTO Seasonality (season_name, start_date, end_date, season_type)
38 VALUES
39 ('Peak Late November', '2025-11-20', '2025-11-30', 'PEAK'),
40 ('Normal Mid-November', '2025-11-10', '2025-11-19', 'NORMAL'),
41 ('Off Peak Early December', '2025-12-01', '2025-12-10', 'OFF_PEAK');
42
43
44 -- -----
45 -- 5. Aircraft
46 -- -----
47 INSERT INTO Aircraft (tail_number, model, manufacture_year, airline_id)
48 VALUES
49 ('HL0001', 'B777', 2015, 1),
50 ('HL0002', 'A330', 2017, 2),
51 ('N12345', 'B787', 2019, 3);
52
53
54 -- -----
55 -- 6. Flight (타켓 2개 + 항공사 평균 3개)
56 -- -----
57
58 -- 타켓 1은 : KE101 (자연 적은 항공사 + Peak + 새벽)
59 INSERT INTO Flight (
60 airline_id, aircraft_id, flight_number,
61 departure_airport, arrival_airport,
62 scheduled_dep, scheduled_arr,
63 estimated_dep, actual_dep, status,
64 season_id
65 ) VALUES (
66 1, 1, 'KE101',
67 1, 2,
68 '2025-11-25 06:00', '2025-11-25 08:20',
69 '2025-11-25 06:10', '2025-11-25 06:05',
70 'DELAYED', 1
71 );
72 SET @F1 = LAST_INSERT_ID();
73
74
75 -- 타켓 2은 : DL202 (자연 큰 항공사 + Off Peak + 오후)
76 INSERT INTO Flight (
77 airline_id, aircraft_id, flight_number,
78 departure_airport, arrival_airport,
79 scheduled_dep, scheduled_arr,
80 estimated_dep, actual_dep, status,
81 season_id
82 ) VALUES (
83 3, 3, 'DL202',
84 1, 3,
85 '2025-12-05 14:00', '2025-12-05 20:00',
86 '2025-12-05 14:40', '2025-12-05 14:45',
87 'DELAYED', 3
88 );
89 SET @F2 = LAST_INSERT_ID();
90
91
92 -- 항공사 평균 계산용 (KE)
93 INSERT INTO Flight (
94 airline_id, aircraft_id, flight_number,
95 departure_airport, arrival_airport, scheduled_dep
96 ) VALUES (1, 1, 'KE201', 1, 2, '2025-11-15 07:00');
97 SET @F3 = LAST_INSERT_ID();
98
99
100 -- 항공사 평균 계산용 (OZ)
101 INSERT INTO Flight (
102 airline_id, aircraft_id, flight_number,
103 departure_airport, arrival_airport, scheduled_dep
104 ) VALUES (2, 2, 'OZ303', 1, 2, '2025-11-16 07:30');
105 SET @F4 = LAST_INSERT_ID();
106
107
108 -- 항공사 평균 계산용 (DL - 자연 선행)
109 INSERT INTO Flight (
110 airline_id, aircraft_id, flight_number,
111 departure_airport, arrival_airport, scheduled_dep
112 ) VALUES (3, 3, 'DL404', 1, 3, '2025-12-06 14:30');
113 SET @F5 = LAST_INSERT_ID();
114
115
116
117 -- -----
118 -- 7. Weather Forecast (타켓 1 = 좋은 날씨 / 타켓 2 = 매우 나쁜 날씨)
119 -- -----
120
121 INSERT INTO Weather_Forecast (
122 airport_id, forecast_datetime, predicted_at,
123 weather_condition, temperature,
124 wind_direction, wind_speed_kt,
125 cloud_ceiling_ft, visibility_km,
126 crosswind_kt, confidence_score, severity_score
127 ) VALUES
128 -- 타켓 1: 날씨 좋음
129 (1, '2025-11-25 06:00', '2025-11-24 22:00',
130 '맑음', 2, 300, 5, 5000, 10, 2, 0.90, 0.1),
131
132 -- 타켓 2: 악천후
133 (1, '2025-12-05 14:00', '2025-12-05 09:00',
134 '폭우', 7, 250, 25, 2000, 4, 15, 0.70, 0.9);
135
136 SET @W1 = 1;
137 SET @W2 = 2;
138
139
```

```

141 -- -----
142 -- 8. Flight = Weather Forecast 航班
143 -- -----
144 INSERT INTO Flight_Weather_Forecast (flight_id, forecast_id)
145 VALUES
146 (@F1, @W1),
147 (@F2, @W2);
148
149
150
151 -- -----
152 -- 9. Flight Delay (항공사별 지연 패턴 분석 반영)
153 -- -----
154
155 -- KE 지연 폭증 (평균 12~15분)
156 INSERT INTO Flight_Delay VALUES
157 (NULL, @F1, 10, 1, '2025-11-25 05:40', 'DEPARTURE'),
158 (NULL, @F1, 15, 1, '2025-11-25 05:50', 'DEPARTURE'),
159 (NULL, @F3, 12, 1, '2025-11-15 06:50', 'DEPARTURE');
160
161 -- OZ 중거리 지연 (평균 30분)
162 INSERT INTO Flight_Delay VALUES
163 (NULL, @F4, 30, 1, '2025-11-16 07:15', 'DEPARTURE');
164
165 -- DL 지연 심화 (평균 55+)
166 INSERT INTO Flight_Delay VALUES
167 (NULL, @F2, 50, 1, '2025-12-05 13:20', 'DEPARTURE'),
168 (NULL, @F2, 55, 1, '2025-12-05 13:30', 'DEPARTURE'),
169 (NULL, @F5, 60, 1, '2025-12-06 14:10', 'DEPARTURE');
170
171
172
173 -- -----
174 -- 10. Airport Congestion
175 -- -----
176 INSERT INTO AirportCongestion VALUES
177 (NULL, 1, '2025-11-25 06:00', 20, 'MEDIUM'),
178 (NULL, 1, '2025-12-05 14:00', 45, 'HIGH');
179
180
181
182 -- -----
183 -- 11. User
184 -- -----
185 INSERT INTO User (name, email)
186 VALUES
187 ('Kim Taeho', 'taeho@example.com'),
188 ('Lee Jisoo', 'jisoo@example.com');
189
190
191 -- -----
192 -- 12. User Ticket
193 -- -----
194 INSERT INTO User_Ticket (user_id, flight_id, booking_date, seat_number)
195 VALUES
196 (1, @F1, '2025-11-01', '12A'),
197 (2, @F2, '2025-11-20', '14C');
198

```

(5) Delay Prediction Test Based on User Ticket

An SQL query was executed that takes the user's ticket (ticket_id) as input and performs an integrated analysis of past delay data under similar conditions. The results confirmed that factors such as airline, weather, season, time of day, and congestion were appropriately reflected in the output.

```

1 SET @ticket_id = 1;
2
3 SELECT
4   @ticket_id AS ticket_id,
5
6   -- 1) 날씨 영향
7   (
8     SELECT ROUND(AVG(fd.delay_minutes), 1)
9     FROM Flight f
10    JOIN Flight_Weather_Forecast fwf ON f.flight_id = fwf.flight_id
11    JOIN Weather_Forecast wf ON fwf.forecast_id = wf.forecast_id
12    JOIN Flight_Delay fd ON f.flight_id = fd.flight_id
13    WHERE f.flight_id = ut.flight_id
14    AND wf.weather_condition = wf_target.weather_condition
15    AND ABS(wf.wind_speed_kt - wf_target.wind_speed_kt) <= 3
16  ) AS weather_delay,
17
18   -- 2) 항공사 영향
19   (
20     SELECT ROUND(AVG(fd.delay_minutes), 1)
21     FROM Flight f2
22    JOIN Flight_Delay fd ON f2.flight_id = fd.flight_id
23    WHERE f2.airline_id = f_target.airline_id
24  ) AS airline_delay,
25
26   -- 3) 시즌 영향
27   (
28     SELECT ROUND(AVG(fd.delay_minutes), 1)
29     FROM Flight f3
30    JOIN Seasonality s ON f3.season_id = s.season_id
31    JOIN Flight_Delay fd ON f3.flight_id = fd.flight_id
32    WHERE f3.season_id = f_target.season_id
33  ) AS season_delay,
34
35   -- 4) 시간대 영향
36   (
37     SELECT ROUND(AVG(fd.delay_minutes), 1)
38     FROM Flight f4
39    JOIN Flight_Delay fd ON f4.flight_id = fd.flight_id
40    WHERE HOUR(f4.scheduled_dep)
41      BETWEEN HOUR(f_target.scheduled_dep) - 1
42      AND HOUR(f_target.scheduled_dep) + 1
43  ) AS time_delay,
44
45   -- 5) 공항도 영향
46   (
47     SELECT ROUND(AVG(fd.delay_minutes), 1)
48     FROM AirportCongestion ac
49    JOIN Flight f5 ON ac.airport_id = f5.departure_airport
50    JOIN Flight_Delay fd ON f5.flight_id = fd.flight_id
51    WHERE f5.departure_airport = f_target.departure_airport
52    AND ac.timeslot = DATE_FORMAT(f_target.scheduled_dep, '%Y-%m-%d %H:00')
53  ) AS congestion_delay,
54
55   (
56     COALESCE(
57       (
58         SELECT ROUND(AVG(fd.delay_minutes), 1)
59         FROM Flight f
60        JOIN Flight_Weather_Forecast fwf ON f.flight_id = fwf.flight_id
61        JOIN Weather_Forecast wf ON fwf.forecast_id = wf.forecast_id
62        JOIN Flight_Delay fd ON f.flight_id = fd.flight_id
63        WHERE f.flight_id = ut.flight_id
64        AND wf.weather_condition = wf_target.weather_condition
65        AND ABS(wf.wind_speed_kt - wf_target.wind_speed_kt) <= 3
66      ), 0)
67     +
68     COALESCE(
69       (
70         SELECT ROUND(AVG(fd.delay_minutes), 1)
71         FROM Flight f2
72        JOIN Flight_Delay fd ON f2.flight_id = fd.flight_id
73        WHERE f2.airline_id = f_target.airline_id
74      ), 0)
75     +
76     COALESCE(
77       (
78         SELECT ROUND(AVG(fd.delay_minutes), 1)
79         FROM Flight f3
80        JOIN Seasonality s ON f3.season_id = s.season_id
81        JOIN Flight_Delay fd ON f3.flight_id = fd.flight_id
82        WHERE f3.season_id = f_target.season_id
83      ), 0)
84     +
85     COALESCE(
86       (
87         SELECT ROUND(AVG(fd.delay_minutes), 1)
88         FROM Flight f4
89        JOIN Flight_Delay fd ON f4.flight_id = fd.flight_id
90        WHERE HOUR(f4.scheduled_dep)
91          BETWEEN HOUR(f_target.scheduled_dep) - 1
92          AND HOUR(f_target.scheduled_dep) + 1
93      ), 0)
94     +
95     COALESCE(
96       (
97         SELECT ROUND(AVG(fd.delay_minutes), 1)
98         FROM AirportCongestion ac
99        JOIN Flight f5 ON ac.airport_id = f5.departure_airport
100       JOIN Flight_Delay fd ON f5.flight_id = fd.flight_id
101       WHERE f5.departure_airport = f_target.departure_airport
102       AND ac.timeslot = DATE_FORMAT(f_target.scheduled_dep, '%Y-%m-%d %H:00')
103     ), 0)
104   ) AS total_delay
105 FROM User_Ticket ut
106 JOIN Flight f_target ON ut.flight_id = f_target.flight_id
107 JOIN Flight_Weather_Forecast fwf_target ON f_target.flight_id = fwf_target.flight_id
108 JOIN Weather_Forecast wf_target ON fwf_target.forecast_id = wf_target.forecast_id
109 WHERE ut.ticket_id = @ticket_id;

```

```

1 • SET @ticket_id = 1;
2
3 • SELECT
4   @ticket_id AS ticket_id,
5
6   -- 1) 날씨 영향
7   (
8     SELECT ROUND(AVG(fd.delay_minutes), 1)
9     FROM Flight f
10    JOIN Flight_Weather_Forecast fwf ON f.flight_id = fwf.flight_id
11    JOIN Weather_Forecast wf ON fwf.forecast_id = wf.forecast_id
12    JOIN Flight_Delay fd ON f.flight_id = fd.flight_id
13    WHERE f.flight_id = ut.flight_id

```

ticket_id	weather_delay	airline_delay	season_delay	time_delay	congestion_delay	total_delay
1	12.5	12.3	12.5	16.8	33.1	87.2

4. Evaluation

The FlightDelayDB designed in this project integrates various factors affecting flight delays into a relational model, enabling trend analysis through SQL-based queries. Test results confirmed that individual factors—such as airline, weather conditions, time of day, congestion, and seasonality—were correctly reflected in the analysis. Additionally, the system functioned as intended in retrieving past delay cases under similar conditions based on the user’s ticket information.

By observing different outcomes in delay prediction based on variations in sample data

conditions, the data model was verified to appropriately represent the relative impact of each delay factor. Integrity constraints such as foreign key and UNIQUE constraints worked as expected, and duplicate prevention in mapping tables was also confirmed. However, the current project was built using sample data for educational purposes and does not include actual airline internal data or air traffic control information. Incorporating these elements in the future would improve the realism and practical applicability of the analysis.

5. Conclusion

This project successfully developed a relational database system that stores key factors of flight delays in an integrated manner and enables analysis of flight delay probability through SQL queries. Centered around flight schedules, the system incorporates weather data, congestion levels, seasonality, airline operational characteristics, and delay records to support multi-factor analysis.

The feature that retrieves past delay cases under similar conditions based on a user's ticket and estimates expected delay time demonstrates the potential to serve as a practical decision-support tool. In the future, integrating real-time weather APIs, aircraft turnaround times, and air traffic control restrictions could enable more precise and real-world applicable delay analysis.

6. Reference

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